

Amazon FBA Stock Health & Early Warning System

A daily forecasting and recommendation product that predicted stock risk across 50k+ FBA seller accounts and told 600+ EU account managers what to restock, how much and by when.

Hrishikesh Choudhari Amazon EU Data & Analytics

600+ account managers using it every week	50k+ FBA seller accounts covered	28% fewer stockouts in peak season	8 hrs saved per account manager each week
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EXECUTIVE SUMMARY

Act on stock risk before it becomes lost sales

CONTEXT	Amazon earns around 35% of the price of every FBA sale, so it only earns when a seller's product is in stock. About 70% of FBA shipments come from sellers manufacturing in China, with roughly 10 days in transit. Restock decisions therefore have to be made well before shelves run low.
PROBLEM	Each of the 600+ account managers tracked hundreds of sellers through manual spreadsheet checks. Stockouts were usually spotted after sales were lost, while slow-moving stock built up storage costs.
SOLUTION	Every morning, the product forecasts demand for each SKU, compares it with stock on hand and in transit, and recommends how many units to send and the latest ship-by date. Accounts are ranked by revenue at risk, so account managers call the most urgent sellers first.
OUTCOME	Became the default weekly planning tool for EU account management. In peak season it cut stockouts by 28% and saved each account manager 8 hours a week.

The balance it optimises

Too little stock

Lost sales for the seller and lost revenue share for Amazon.

Too much stock

Storage fees for the seller and wasted warehouse space for Amazon.

The right amount

Demand met at the lowest cost for both.

Who uses it, and for what

Account managers

Which sellers to call this week, and what to advise.

Sellers

How many units to send, and by when.

Leadership

Where stock risk sits across EU marketplaces.

Forecast demand → Find the gap → Recommend an order → Account manager acts → Model learns

What I owned, from requirements to adoption

DEFINE - Gathered functional and non-functional requirements with account managers and vendors - Designed the technical architecture and the dataset metadata - Secured Privacy team approval to process vendor data, confirming no customer PII, performance or safety data was involved	BUILD - Orchestrated delivery across data engineers (infrastructure) and data scientists (forecasting model) - Built the ETL in Databricks and the account manager dashboard myself - Designed automated row-level security, so each account manager sees only their own sellers	LAUNCH - Led the launch and drove adoption across 600+ account managers - Ran regular office hours to support users and gather feedback - Owned the roadmap of improvements after launch
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METADATA

Output dataset

The governed table that every dashboard, alert and export reads from.

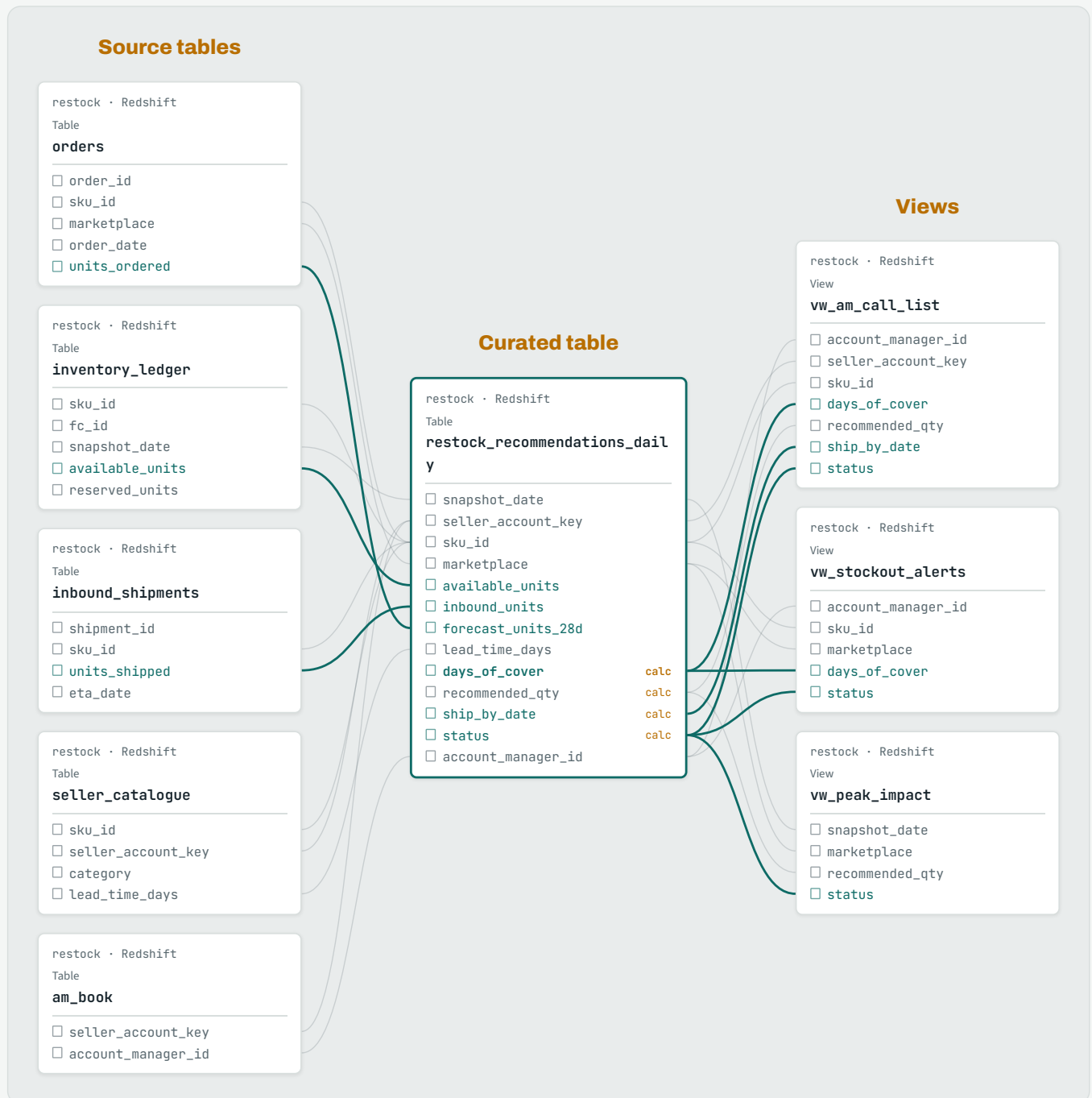
restock_recommendations_daily			
OWNER EU Data & Analytics (product owner: Hrishikesh Choudhari)	CONSUMERS 600+ EU account managers, ops leadership	GRAIN Seller account × SKU × marketplace × day	REFRESH AND SLA Daily batch, available by 07:00 CET
RETENTION 24 months, for year-on-year seasonality	CLASSIFICATION Confidential. No customer PII; seller IDs hashed	ACCESS Row-level: each AM sees only their own book of accounts	QUALITY CHECKS Freshness, row-count drift, null keys, stock reconciliation

COLUMN	TYPE	DESCRIPTION
snapshot_date	DATE	Date the recommendation was generated
seller_account_key	STRING	Hashed seller account identifier
sku_id	STRING	Seller SKU
marketplace	STRING	UK, DE, FR, IT, ES and other EU stores
category	STRING	Product category
available_units	INT	Sellable units in fulfilment centres
inbound_units	INT	Units shipped but not yet received
forecast_units_28d	DECIMAL	Expected demand, next 28 days (mean)
forecast_p90_28d	DECIMAL	High-demand scenario used for safety stock
lead_time_days	INT	Seller's typical time from order to stock received
days_of_cover	DECIMAL	How long current stock plus inbound will last
recommended_qty	INT	Units to send; 0 when stock is healthy
ship_by_date	DATE	Latest date to ship and avoid a stockout
status	ENUM	critical, reorder, healthy, excess

COLUMN	TYPE	DESCRIPTION
priority_score	DECIMAL	Ranks accounts by revenue at risk, for the AM call list
account_manager_id	STRING	Owner of the seller relationship; drives row-level access
SOURCE DATASET	GRAIN	REFRESH
Customer orders	Order line	Hourly, aggregated daily
Inventory ledger	SKU × fulfilment centre snapshot	Daily
Inbound shipments	Shipment line	Daily
Seller and catalogue master	SKU attributes, lead times	Daily
Account manager book	AM to seller mapping	Weekly
Fees and storage limits	Seller × storage type	Monthly

Column-level lineage

Every field an account manager sees traces back to its source column. The highlighted path shows how `days_of_cover` is built from three source tables.

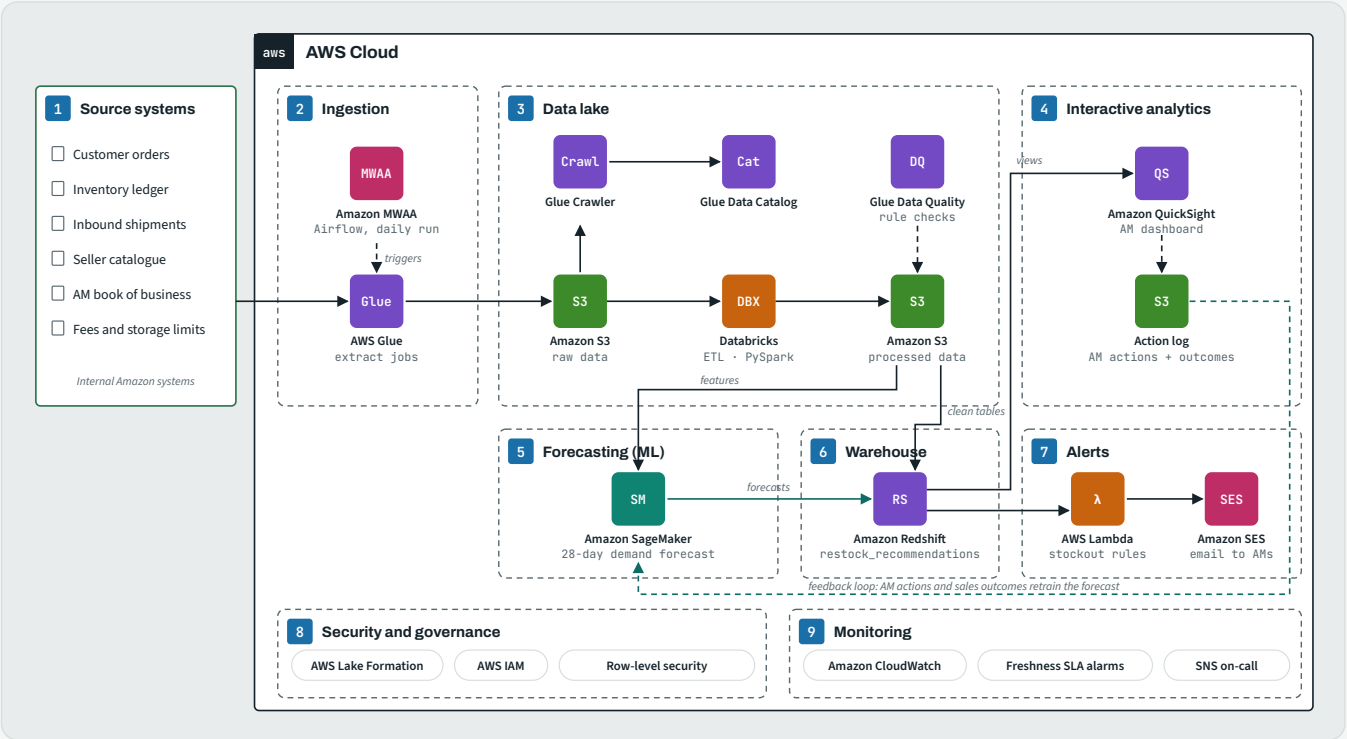


Tracing **days_of_cover** · from 3 source columns · used in vw_am_call_list, vw_stockout_alerts, vw_peak_impact

Feedback loop · Whether AMs actioned each recommendation, and what sold afterwards, flows back into the model to track acceptance, forecast accuracy and drift.

How it runs on AWS

A daily batch pipeline: land, clean, forecast, recommend, serve, then learn from what account managers did.



1. **Source systems** export orders, stock, inbound shipments, catalogue, AM ownership and fees.

2. **Ingestion:** Airflow (MWAA) triggers Glue extract jobs every night.

3. **Data lake:** raw files land in S3, crawlers register them in the Data Catalog, Databricks ETL cleans and joins them, and quality rules gate what moves on.

4. **Interactive analytics:** QuickSight serves the AM dashboard with row-level security; every action is logged.

5. **Forecasting:** SageMaker produces a 28-day demand forecast (mean and P90) per SKU and marketplace.
6. **Warehouse:** Redshift combines forecasts with stock and lead times into `restock_recommendations_daily`.

7. **Alerts:** Lambda applies stockout rules and emails each AM their critical accounts through SES.

8. **Security and governance:** Lake Formation and IAM control access; seller IDs are hashed and no customer PII is used.

9. **Monitoring:** CloudWatch tracks freshness against the 07:00 CET SLA and pages on-call through SNS.

METRICS

Metric catalogue

One definition per metric, held in the curated layer so every view agrees.

METRIC	DEFINITION	WHY IT MATTERS
Days of cover	$(\text{available} + \text{inbound}) \div \text{avg daily forecast}$	Core health signal for every SKU
Stockout risk	$\text{days of cover} < \text{lead time} + \text{safety buffer}$	Triggers critical and reorder status
Recommended qty	$\text{forecast over} (\text{lead time} + \text{coverage}) + \text{safety stock} - (\text{available} + \text{inbound})$	Turns insight into an action
Safety stock	$\text{P90 demand} - \text{mean demand over lead time}$	Protects against demand spikes
Sell-through rate	$\text{units sold} \div (\text{units sold} + \text{ending stock})$	Shows how fast stock moves
Excess inventory	units above 90 days of cover	Flags storage cost and removal candidates
In-stock rate	% of days a SKU was sellable	Main outcome metric for sellers

Metric	Definition	Why it matters
Forecast accuracy	1 - WAPE, plus bias	Keeps trust in the recommendations
Acceptance rate	recs actioned within 7 days ÷ recs issued	Adoption: does the product change behaviour?

DASHBOARD

Account manager view

Illustrative values. Opens on the AM's own book, sorted by revenue at risk.

My accounts · this week

Marketplace: All EUCategory: AllStatus: Critical + Reorder

Accounts at stockout risk

14

+3 vs last week

Median days of cover

34

+2 days

Recommendations actioned

62%

+5 pts

Forecast accuracy (1 - WAPE)

81%

steady

Weekly demand: forecast vs actual

Stock status across my book

Critical14

Reorder38

Healthy212

Excess29

SKUs in book: 293

SELLER · SKU	MKT	DAYS OF COVER	INBOUND	SEND	SHIP BY	STATUS
Seller A · Kitchen kettle	DE	4	0	1,240	Today	Critical
Seller E · Yoga mat	UK	6	500	420	+2 days	Critical
Seller B · Puzzle set	FR	11	300	860	+5 days	Reorder
Seller C · Face serum	IT	27	0	0	-	Healthy
Seller D · Throw cushion	ES	96	0	0	Remove 400	Excess

What changed

Measured over peak season, when demand spikes and a stockout costs the most.

ADOPTION

600+ AMs

The default weekly planning tool for EU account management, across 50k+ seller accounts.

AVAILABILITY

28% fewer stockouts

Risk flagged early enough for sellers to ship stock before running out.

EFFICIENCY

8 hrs saved per AM / week

A ranked call list replaced manual spreadsheet checks, freeing roughly 4,800 hours a week across the team for seller conversations.

Every stockout avoided is a completed sale, and Amazon keeps around 35% of its price.

RELEVANCE TO VOCALINK

The same pattern, applied to payments

IN THE STOCK HEALTH & EARLY WARNING SYSTEM	AT VOCALINK
Forecast demand before stock runs out	Predict payment failures or returns before they reach the customer
Recommend a quantity and a ship-by date	Give banks a risk score plus a recommended action
Row-level access to each AM's own book	Each bank sees its own data, plus anonymised market benchmarks
Acceptance rate and forecast accuracy feedback loop	Monitor model drift and adoption of fraud and scam scores
Revenue protected through the cost-share model	A clear value case for each bank, which supports pricing and adoption